Math Comparative Advantage and the Gender Gap in STEM: From Upper-Secondary Tracks to University Degrees

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Abstract

Even among top-decile mathematics students entering Catalan universities, 28% of girls enroll in STEM degrees, against 60% of boys. How much of women's underrepresentation in STEM does math comparative advantage (measured as the difference between standardized mathematics and reading scores) account for? Administrative records follow 444,033 students through two decisions: the Science track among upper-secondary students and a STEM degree among university entrants. Conditional on school, cohort, and family background, 10th-grade comparative advantage accounts for about half of the 16.3-percentage-point Science-track gap but only a fifth of the 28.6-point STEM-degree gap, mainly because that gap is larger; counting life sciences and medicine, the share nears two-fifths. Much of this overlaps with mathematics achievement, beyond which peer rank adds little; the shares are lower bounds under classical measurement error. At university, girls and boys with equal mathematics and reading scores still differ by 22.8 points.

Keywords: Gender gap; STEM; comparative advantage; peer rank; track choice; field of study; mathematics achievement.

JEL Codes: J16, J24, I21, I23, I24.

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1 Introduction

Women have made substantial gains in educational attainment (Becker et al., 2010; Klasing and Milionis, 2020), yet remain markedly underrepresented in science, technology, engineering, and mathematics (STEM). They account for only about 35% of STEM graduates worldwide, a share that has remained virtually unchanged over the past decade, with particularly pronounced underrepresentation in engineering and information and communication technology (UNESCO, 2021; Global Education Monitoring Report Team, 2024). Average gender differences in mathematics achievement at school age, however, are small in most education systems (Breda and Napp, 2019; Borgonovi et al., 2021), so absolute mathematics achievement cannot account for the whole of the gender gap in STEM choices. We ask how much of that gap is accounted for by mathematics achievement, and how much by what that achievement is compared with.

We focus on two comparisons of a student's mathematics achievement that the literature on academic self-evaluation emphasizes (Möller et al., 2009). The first is with the student's own achievement in other subjects, an internal comparison across subjects and the educational analogue of comparative advantage in Roy-type models of self-selection (Roy, 1951; Willis and Rosen, 1979). The second is with the mathematics achievement of peers in the same school cohort, an external comparison that gives the student's peer rank in mathematics. The two need not coincide. A student may rank highly in mathematics among their peers while performing even better in reading, whereas another may have mathematics as their strongest subject despite occupying a more moderate position in the mathematics distribution. We do not observe students' subjective perceptions of their performance. Rather, we study observable academic profiles corresponding to these two comparisons and ask how much of the gender differences in STEM specialization they account for, on their own and beyond absolute mathematics achievement.

Breda and Napp (2019) brought cross-subject academic profiles to the center of the STEM gender-gap debate, showing that girls' mathematics-minus-reading differential accounts for most of the gender gap in intentions to pursue mathematics-intensive studies and careers. Later work finds smaller and more heterogeneous shares, for open-ended intentions as well as realized choices (Wan et al., 2023). In upper-secondary education, the comparison accounts for a modest part of the gender gap in Sweden (Jonsson, 1999), about half in a Paris-region sample (Breda and Napp, 2019), and more than half of the unexplained gender difference in choosing advanced STEM exams in Poland (Sadowski and Zawistowska, 2020), while in Italy it barely changes the gap (Contini et al., 2025). At university, it accounts for a small part of the gap in the United States and Ireland (Riegle-Crumb et al., 2012; Delaney and Devereux, 2019), a quarter to a third in England (Aucejo and James, 2021), and most of it among STEM-ready entrants in Ontario (Card and Payne, 2021); in Sweden, it accounts for a small part of the gap in the orientation of later education and occupations (Dekhtyar et al., 2018).

Rank among peers is associated with later educational choices even conditional on absolute achievement: a high primary-school rank in a subject raises the probability of taking it at A-

level in England (Murphy and Weinhardt, 2020), class rank shapes track choice in Italy (Contini et al., 2025), and rank is associated with STEM applications in Ireland (Delaney and Devereux, 2021) and, measured on overall grades, with the choice of math-intensive electives at a Dutch business school (Elsner et al., 2021). Where it has been measured, rank accounts for only a small part of the gender gap beyond absolute achievement (Murphy and Weinhardt, 2020; Delaney and Devereux, 2021; Contini et al., 2025). Other work ranks a within-student ratio among classmates or combines ranks across subjects (Goulas et al., 2024; Fuchs et al., 2025; Liu, 2025), without reporting how much of the gender gap a separately measured peer rank in mathematics accounts for. Because rank is built from the mathematics score, how much of what it accounts for on its own is shared with that score remains to be shown.

The shares reported in these studies are not estimates of a common parameter: studies also differ in the outcome, the construction of the comparison, the definition of STEM, the baseline gap (Card and Payne, 2021), and the age at which performance is measured, although in Germany the same tests yield a similar share at different ages (Uunk, 2024). What remains open is how within-student and peer-relative measures, constructed separately, compare across successive specialization margins in one setting, with the same data and measures.

We address these questions by examining two successive specialization decisions using linked administrative data from Catalonia (Spain). The first occurs around age 16, when students entering upper-secondary education (*Batxillerat*) choose among several tracks, including the Science and Technology track (hereafter the Science track), which focuses on mathematics and the natural sciences. The second occurs upon entry into university, when students enroll in a degree program. To our knowledge, no study relates within-student math comparative advantage, measured for the same students at the end of primary school and at the end of compulsory secondary school, to two successive realized specialization decisions, alongside a separately measured peer rank and absolute mathematics achievement. The studies closest to ours examine a single track-choice margin (Contini et al., 2025) or follow students to university application with one measure combining cross-subject and peer-relative performance (Goulas et al., 2024). Holding the setting, data, and measures fixed, we compare the accounting contribution of each measure across margins and, for achievement and comparative advantage, across ages. We also distinguish whether a measure accounts for more of the gender gap because enrollment is more strongly associated with it or because girls and boys differ more in it, and assess how much of the information in comparative advantage and peer rank overlaps with absolute mathematics achievement.

We link 6th- and 10th-grade assessments to subsequent enrollment for 444,033 students of the 2000–2007 birth cohorts, of whom 317,555 reach *Batxillerat* and, in the 2000–2005 cohorts, 147,732 go on from it to a Catalan university. We measure comparative advantage in mathematics as the difference between standardized mathematics and reading scores and additionally construct students' peer-relative mathematics rank. At each educational margin, we estimate the gender gap conditional on school, cohort, and student and family characteristics. The share

accounted for by a performance measure is the percentage reduction in this gap when the measure is added. These exercises are descriptive rather than causal: they characterize how observed academic profiles are associated with the gender gap among students reaching each stage.

Our first result is that the share of the gender gap accounted for by math comparative advantage varies substantially across educational margins and with the definition of STEM. Among students observed in *Batxillerat*, 10th-grade comparative advantage accounts for 49.4% of the gender gap in Science-track enrollment. Among university entrants, it accounts for 20.0% of the gap in STEM-degree enrollment under the canonical STEM definition, which covers the physical sciences, mathematics, computing, and engineering. Most of that gap arises among students who followed the Science track; differences in which track girls and boys followed account for only about 28% of it. When life sciences and medicine are also classified as STEM, the outcome closest to what the Science track prepares students for, the gap is smaller and the share rises to 39.1%, against 49.4% in the Science track, because girls in the Science track frequently enter health-related programs, whereas boys rarely do. The contrast between the two margins comes mainly from the larger STEM-degree gap: in percentage points, the two reductions are much closer. Timing also matters: comparative advantage measured in 6th grade accounts for much less of both gaps than the 10th-grade measure, though the comparison cannot say why, since the two assessments differ and the later one is closer to the decisions.

Second, among students who reach each stage, math comparative advantage accounts for more of the gap than mathematics achievement because it adds girls' reading advantage to their mathematics disadvantage; on the observed scores, enrollment does not rise more steeply with it. Entered jointly, the two overlap heavily: comparative advantage adds less than four percentage points to the share accounted for by mathematics achievement, and the Gelbach (2016) decomposition assigns it 5.2% of the Science-track gap and 6.0% of the STEM-degree gap. Because this split depends on the pair of measures entered and on measurement error, we read it as evidence of overlap with mathematics achievement rather than as an independent comparative-advantage component.

Third, peer-relative mathematics rank is positively associated with STEM specialization conditional on mathematics achievement. Entered alone, it accounts for about as much of each gap as mathematics achievement, but adding it to mathematics achievement raises the share of the gap accounted for by less than one percentage point at both margins. This small increment does not imply that relative standing is unrelated to educational choices: in these data, much of the part of the gender gap associated with peer rank overlaps with the mathematics achievement from which rank is constructed.

Fourth, the gap that remains looks very different across the two margins in percentage points, and much less so on a latent-index scale. In the Science track, girls' and boys' enrollment rises at similar rates with math comparative advantage, and the gap changes relatively little across quartiles of 10th-grade mathematics. At university, girls' enrollment rises a little more than half

as steeply as boys' (slopes of 0.083 and 0.144), and the gap rises from about 10 percentage points in the bottom quartile to more than 30 points in the top. On a latent-index scale, girls' slope on math comparative advantage is, if anything, the steeper, and the STEM-degree gap changes little across quartiles. We therefore read the widening descriptively, as a large gap among the strongest mathematics students; the widening in stated intentions documented by Breda et al. (2023), by contrast, survives logit and probit models.

Taken together, the findings caution against treating math comparative advantage as a general account of women's underrepresentation in STEM. With mathematics achievement, it accounts for nearly half of the gender gap at the choice of track but only about a fifth at university, where girls and boys with the same mathematics and reading scores still differ by 22.8 percentage points. Its share of the gap is not a stable property of the measure: it varies with the educational margin, the definition and size of the underlying STEM gap, the stage at which performance is measured, the way comparative advantage is constructed, and whether the sample conditions on continuing in education, and much of it overlaps with absolute mathematics achievement. Estimates of that share can therefore differ substantially across studies, and should not by themselves be read as evidence for a distinct behavioral mechanism. The results point to where explanations beyond academic profiles are most needed: the choice of degree among students with the same scores.

The rest of the paper proceeds as follows. Section 2 describes the institutional setting and data. Section 3 presents the empirical framework and the interpretation of the accounting exercises. Section 4 reports the results, and Section 5 concludes.

2 Context and Data

In Catalonia, students entering academic upper-secondary education (*Batxillerat*, grades 11–12) choose a two-year track around age 16: Arts, Humanities and Social Sciences, or Science and Technology.[1] The Science track includes both STEM-oriented and health-related curricula. Track choice determines coursework and strongly shapes access to university programs.

University entry makes this early specialization consequential: in Catalonia, as in the rest of Spain and unlike in the United States, students apply directly to specific degree programs and enroll in a field of study from the outset. Access to many STEM degree programs is tightly linked to completing the Science track. The track is not a formal requirement, but applicants are ranked by an admission score that adds optional subject exams, weighted by their relevance to each program, so an engineering program rewards mathematics, a subject taught in the Science track. Consistent with this, very few students enter STEM degrees from other tracks: in our data, 96.4% of the students observed in *Batxillerat* who enter a STEM degree had followed the Science track.

[1] A General track, available in a limited number of Catalan schools since 2022–23, was chosen by almost none of the students observed in *Batxillerat*; they count as not enrolled in the Science track.

2.1 Data

We use administrative records from the Statistical Institute of Catalonia (Idescat) for students of the 2000–2007 birth cohorts who sat the 6th-grade assessment. The records link each student's test scores, school, family background, and enrollment in *Batxillerat* and university. The Catalan education authorities administer standardized competency assessments to students in public and private schools at the end of 6th and 10th grade.[2] We use mathematics, Catalan, and Spanish scores (Catalonia is a bilingual region), plus 10th-grade science scores.[3] Reading is the average of Catalan and Spanish scores; using the higher score is a robustness check. The tests do not determine progression or graduation, although teachers may consider them when grading. The 10th-grade assessment precedes track choice by a few months.

We study students at the end of compulsory education: the 444,033 students with mathematics, Catalan, and Spanish scores in both 6th and 10th grade and complete gender and family-background information. We follow these students through two later decisions, which define our two enrollment samples. Track choice is observed for the 317,555 who go on to *Batxillerat*, 71.5% of them.[4] Degree choice is observed for university entrants, with one limit: university records run through 2023–24 and cover only the 2000–2005 birth cohorts, the 2004 cohort in its first two years of possible entry and the 2005 cohort in its first. Among their 234,370 *Batxillerat* students, 147,732 enter a Catalan university.[5]

We use three nested STEM definitions (see Appendix Table A.1 for details): a canonical definition includes environmental sciences, chemistry, earth sciences, physics, mathematics, statistics, computing, engineering, and architecture. A second, broader definition, the one that also includes life sciences and medicine, adds to the canonical definition the other fields the Science track prepares students for: the life sciences, health programs including medicine, agriculture, forestry, and veterinary science. The math-intensive definition retains the programs with the heaviest mathematics content.

We standardize scores within subject, grade, and cohort among students at the end of compulsory education because the means and dispersions of the original 0–100 scores vary across cohorts. The enrollment samples retain this standardization. Besides the scores themselves, we use two measures built from them: within-student comparative advantage in mathematics (relative to

[2]The assessments are the *Proves de competències bàsiques a sisè de primària* (6th grade) and *a quart d'ESO* (10th grade), designed by the Consell Superior d'Avaluació del Sistema Educatiu, the evaluation agency of the Catalan education authorities. The Catalan evaluation authorities indicate that the 10th-grade test changed framework in 2020, with the start of the *Proves de final d'etapa*.

[3]The extract lacks the 10th-grade science scores of the 2000 birth cohort, except for the 8% of its students who reached 10th grade late, a selected group; we leave that cohort out of every analysis that uses science.

[4]Enrollment in vocational education and training is not observed in our data. In the register of all students in Catalan schools, *Batxillerat* is the main route after compulsory schooling: among students of the 2000–2007 birth cohorts who started post-compulsory education at 16, 77% entered *Batxillerat* and the rest intermediate vocational or arts programs (Registre d'alumnes de Catalunya, Departament d'Educació; authors' calculations).

[5]Enrollment outside Catalonia is rare: of the students who qualified for university admission in Catalonia in 2020–21 and 2021–22 and enrolled in an in-person university, about 98% did so in Catalonia (Ministerio de Universidades, *Datos y cifras del Sistema Universitario Español*, 2021–2022 and 2022–2023 editions).

reading) and peer rank in mathematics.

Our main measure of comparative advantage is $z_M - z_R$, the difference between standardized mathematics z_M and reading z_R scores, as in Sadowski and Zawistowska (2020); Breda and Napp (2019) standardize instead the difference between the raw scores. As robustness checks, we use their construction, z_{M-R}, and the difference between the within-cohort percentile ranks of the two scores. The standard deviation of $z_M - z_R$ is 0.85 in 6th grade and 0.88 in 10th grade. Because both scores are standardized over the whole cohort, the measure compares a student across subjects, not with schoolmates. "Comparative advantage" refers to this measure unless stated otherwise.[6]

For peer rank, we measure 10th-grade mathematics rank among students observed in the same school and cohort as $(r-1)/(n-1)$, where r is the student's position and n the cell size, and standardize it for estimation (Elsner and Isphording, 2017; Murphy and Weinhardt, 2020). We also construct same-gender rank. To limit imprecision, both measures require school-cohort cells with more than 20 students and at least 10 of the student's gender. These restrictions yield 424,479 students, 95.6% of those at the end of compulsory education.[7] We do not construct 6th-grade ranks because the roughly 2,340 primary schools produce smaller cells than the 1,164 secondary schools.

Although enrollment samples are positively selected on achievement, their average math comparative advantage hardly changes. Students observed in *Batxillerat* score 0.2 to 0.3 standard deviations higher in mathematics and reading than all students at the end of compulsory education, and university entrants about 0.2 standard deviations higher than students observed in *Batxillerat* in the same cohorts; math comparative advantage differs by about 0.05 or less between each enrollment sample and the sample it is drawn from.[8]

2.2 Gender Differences in Enrollment and Performance

More girls than boys continue to *Batxillerat* (77.7% versus 65.1%), but once there, girls are 16.7 percentage points less likely than boys to enroll in the Science track, before any controls. Among university entrants, the raw gap in STEM-degree enrollment is 28.7 points under the

[6]The raw-score construction, z_{M-R}, weights the standardized mathematics score by the ratio of the dispersions of the two raw scores: 1.74 in 10th grade, between 1.52 and 2.05 cohort by cohort. That ratio is a property of the assessments, not of students, and it makes z_{M-R} close to a second measure of mathematics achievement: in 10th grade it correlates 0.81 with the standardized mathematics score and 0.04 with the reading score. The percentile-rank difference depends only on the order of students on each test, not on how either test is scaled. Section 4 reports the results with both measures.

[7]Tied scores receive their average position. The first floor alone leaves 427,801 students; the second excludes 3,322 more, so that every column of a table that uses the ranks is estimated on the same students.

[8]Selection nonetheless differs by gender: conditional on school, cohort, and family background, girls' reading advantage shrinks from 0.38 standard deviations among all students to 0.24 in *Batxillerat* and 0.19 among university entrants, while their mathematics disadvantage grows from 0.24 to 0.34, so the gender difference in comparative advantage falls from 0.62 to 0.53 and rests increasingly on mathematics (Appendix Tables B.5 and C.1).

canonical definition (Figure 1, top panel), although this gap is smaller under the definition that also includes life sciences and medicine. The bottom panel of Figure 1 traces specialization: among *Batxillerat* students of the 2000–2005 birth cohorts, 40% of girls and 57% of boys choose the Science track; 8% of girls and 25% of boys enter a STEM degree, almost all of them from that track, and 9% of girls but 3% of boys enter medicine or another health program. Thus only one in five girls who choose the Science track enters a STEM degree, against two in five boys.

In 10th grade, boys score 0.25 standard deviations higher than girls in mathematics and girls score 0.37 standard deviations higher in reading, so the two differences add up: girls' math comparative advantage is 0.62 below boys'. In science, boys score 0.10 standard deviations higher, less than half the gap in mathematics (see Appendix Table A.2 for further gender differences).

10th-grade math comparative advantage differs by gender even within mathematics deciles: at similar mathematics scores, girls score higher in reading, which lowers their math comparative advantage (Figure 2). As a result, average comparative advantage turns positive in the fourth decile for boys but only in the eighth for girls, so from the fourth to the seventh decile the average boy's relative strength is already mathematics while the average girl's is still reading. Science-track enrollment is lower among girls in every decile and exceeds 50% only from the eighth decile, against the seventh for boys; among university entrants in the top decile, 28% of girls enroll in a STEM degree, against 60% of boys. From the fourth to the seventh decile, boys enroll in the Science track 8 to 10 percentage points more often than girls and, among university entrants, in a STEM degree 14 to 24 points more often; in the top three deciles, where girls' average comparative advantage is also in mathematics, the STEM gap is wider still.

The gender difference in math comparative advantage grows from 0.39 in 6th grade to 0.62 in 10th grade, consistent with the age-related divergence in numeracy and literacy gaps documented by Borgonovi et al. (2021). Some of the increase may reflect assessment differences, including the concentration of 6th-grade mathematics scores near the ceiling. Comparative advantage is also less persistent across grades than either score: its correlation is 0.39, versus 0.67 for mathematics and 0.71 for reading (Appendix Table A.3). We therefore use 10th-grade measures as our main predictors and examine measurement-error sensitivity in Section 3.

Gender differences in peer rank are smaller than in math comparative advantage. The normalized gender difference is 0.26 for rank and 0.25 for mathematics achievement, versus 0.75 for math comparative advantage (Appendix Table A.4). Girls' average rank is 0.46, compared with boys' 0.54; the share of girls falls from 60.7% in the lowest rank decile to 39.7% in the highest (Appendix Figure A.1). Rank and comparative advantage correlate at 0.435 and classify 32.9% of students differently: their rank is above the median of their school cohort but their comparative advantage favors reading, or vice versa. The first combination is almost twice as common among girls as boys (21.1% versus 11.8%; Appendix Table A.5).

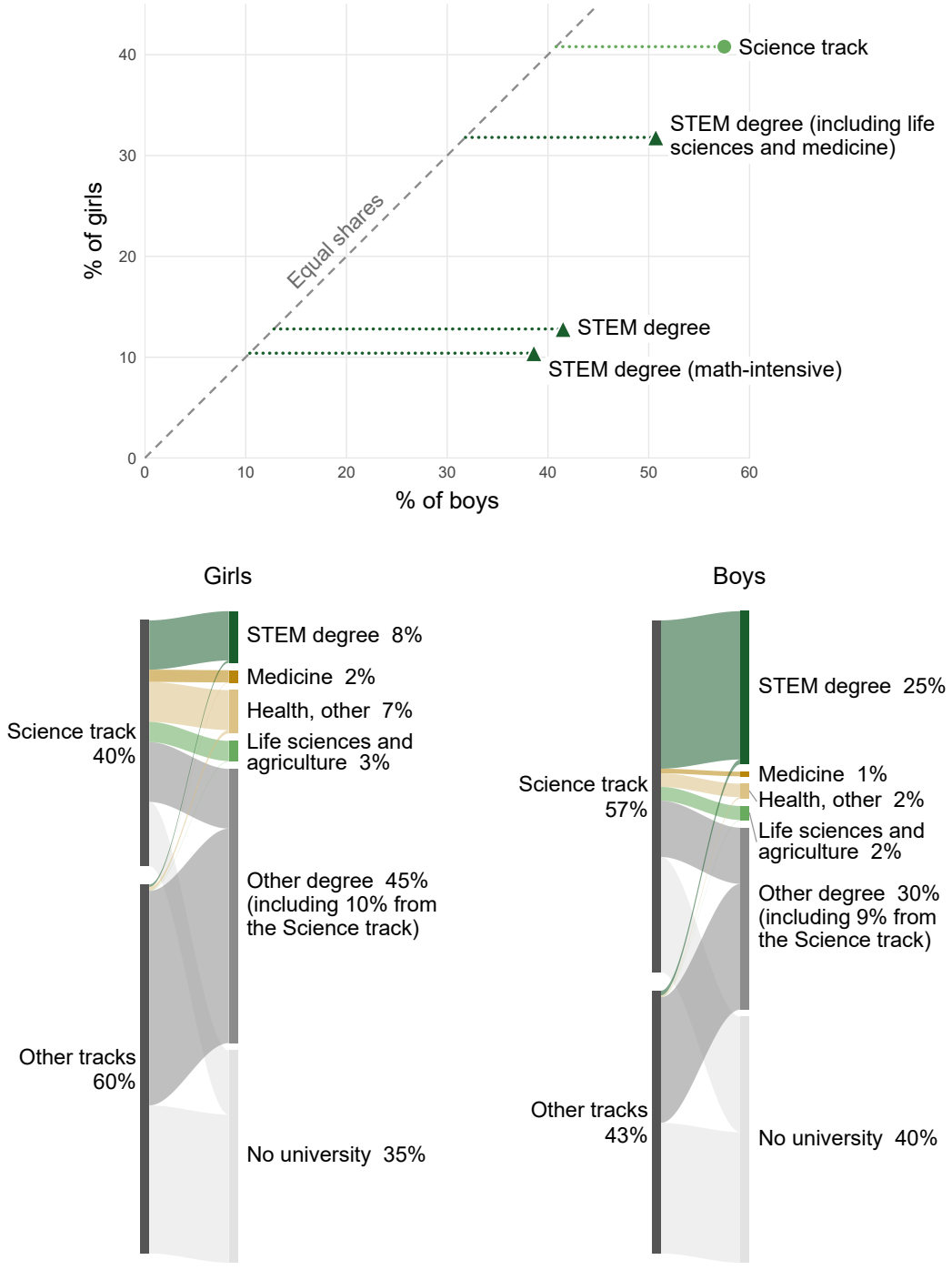


Figure 1: **STEM enrollment of girls and boys, by stage.** Top panel: share of each gender enrolled in each outcome, among students observed in *Batxillerat* for the Science track (N = 317,555) and among university entrants of the 2000–2005 birth cohorts observed in *Batxillerat* for the STEM degrees (N = 147,732). Bottom panel: flows from *Batxillerat* track to university destination, as a percent of each gender's *Batxillerat* students of the 2000–2005 birth cohorts; destinations are mutually exclusive and exhaustive. STEM uses the canonical definition. Source: Idescat administrative records; authors' calculations.

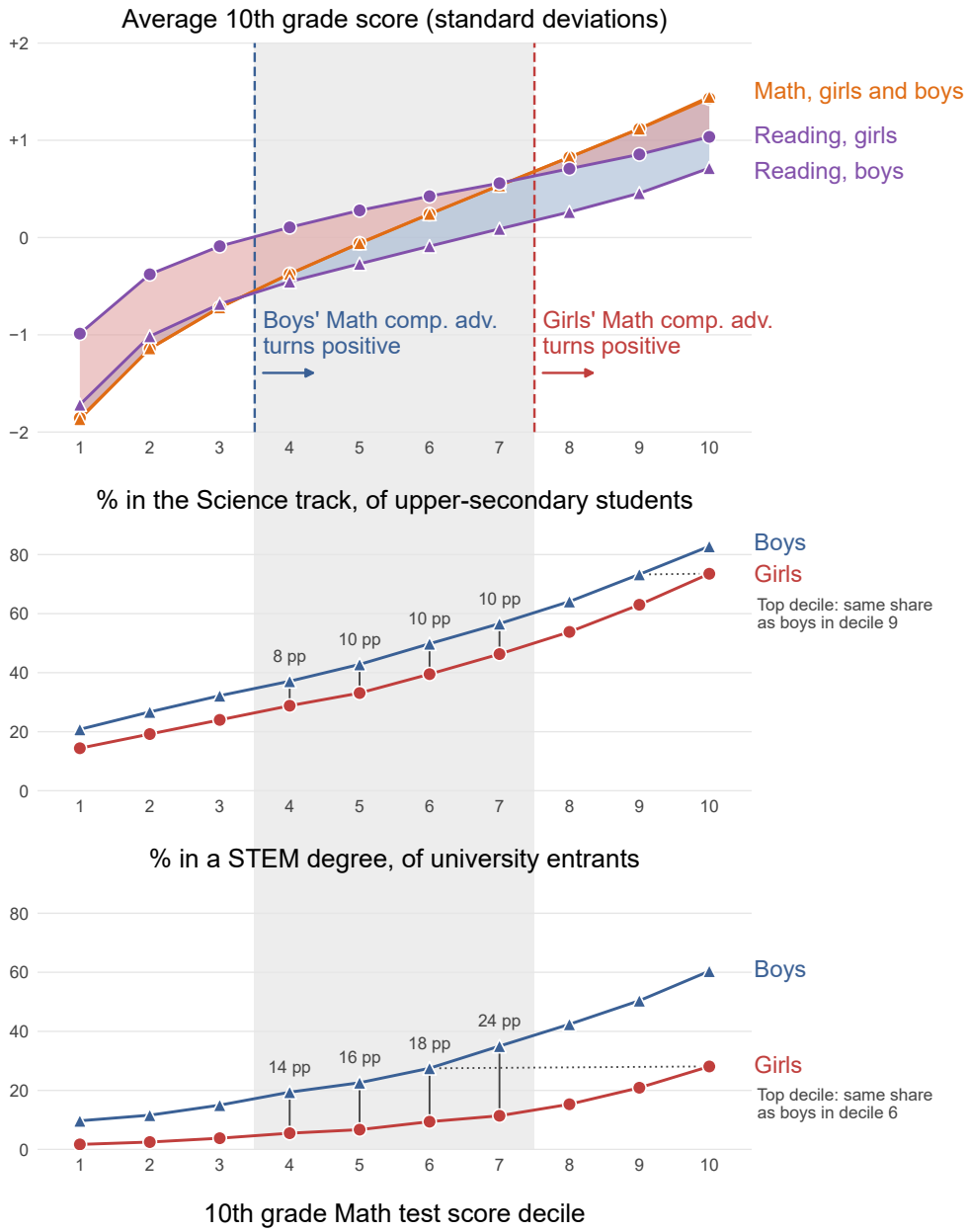


Figure 2: **10th-grade academic profiles and STEM enrollment by math decile, by gender.** By decile of the 10th-grade math score. Top panel: average standardized math and reading scores of girls and boys at the end of compulsory education (N = 444,033); the shaded areas show each gender's gap between the two scores, and the dashed lines mark where each gender's average math comparative advantage turns positive. Middle panel: share of the students observed in *Batxillerat* who enroll in the Science track (N = 317,555). Bottom panel: share of the university entrants of the 2000–2005 birth cohorts observed in *Batxillerat* who enroll in a STEM degree, canonical definition (N = 147,732). The two lower panels share one scale. Gray band: deciles in which, among all students, girls' average math comparative advantage is negative and boys' positive; the numbers in it give the gap between boys and girls in percentage points. Dotted lines join girls' top decile to the boys' decile with the closest share. Source: Idescat administrative records; authors' calculations.

3 Empirical Framework

We relate STEM specialization to students' academic profiles with linear probability models, estimated separately for each choice:

$$Y_{isc} = \alpha\, Female_i + \beta A_i + X_i'\delta + \lambda_s + \gamma_c + \varepsilon_{isc}. \tag{1}$$

Here, Y_{isc} indicates Science-track enrollment among *Batxillerat* students or STEM-degree enrollment. $Female_i$ identifies girls, and A_i is one performance measure at a time: the mathematics score, reading score, or math comparative advantage in 6th or 10th grade, the 10th-grade science score, or peer rank in mathematics (among all or same-gender schoolmates). Controls X_i include student and parental foreign birth, household structure, and parental university education (Appendix Table A.2).[9] School and cohort fixed effects are denoted by λ_s and γ_c; standard errors are clustered by 10th-grade school. The coefficient α is the gender gap, and β the enrollment slope on performance. Both are descriptive because unobserved preferences, beliefs, and other determinants may correlate with gender and achievement.

We use additive school and cohort fixed effects because school-by-cohort effects would identify the gap only within cells containing both genders: among university entrants, a fifth of these cells contain fewer than 10 students and 5.1% contain only one gender. Clustering by school allows shocks to persist across cohorts, and clustering by both the 6th- and the 10th-grade school changes the standard error of the coefficient on girls in the model with comparative advantage by at most 0.0001.

Because our question is what students choose to study rather than whether they continue studying, we estimate each model among the students who reach the corresponding stage. Outcomes defined for all students at the end of compulsory education combine continuation and field choice, which work in opposite directions for girls: they are more likely to enter *Batxillerat* but less likely to choose the Science track once there. Appendix C reports the same models without conditioning on enrollment; there the Science-track gap is 5.5 points, comparative advantage accounts for 87.3% of it and mathematics achievement for 89.5%, so conditioning changes both the size of the shares and the ordering of the two measures.

Because math comparative advantage is the difference between the mathematics and reading scores, the three measures cannot all enter one model, so we first enter them one at a time; comparative advantage entered alone imposes equal and opposite slopes on the two scores. For each measure A_i, the quantity of interest is the share of the gender gap it accounts for, $100(\alpha_0-\alpha_A)/\alpha_0$, where α_0 and α_A are the coefficients on $Female_i$ without and with A_i, estimated on the same students. The reduction satisfies the accounting identity $\alpha_0-\alpha_A = \beta_A\theta_A$, where β_A is the coefficient on A_i and θ_A the conditional gender difference in the measure: the coefficient

[9] Year of birth is omitted because it coincides with the cohort, the year of the Idescat file, for 96.1% of students, and whether a school is publicly or privately owned is absorbed by the school fixed effects.

on $Female_i$ in an auxiliary regression of A_i on $Female_i$, the controls, and the fixed effects, on the same students. A measure on which girls score lower therefore accounts for more of the gap when enrollment rises more steeply with it, when girls and boys differ more in it, or when the gap itself is smaller. This accounting exercise does not identify mediation or the gap reduction that would follow an intervention on academic profiles.

Appendix Tables B.1 to B.4 report the estimates for each outcome, and Appendix Table B.5 the conditional gender differences and the resulting shares. For 10th-grade math comparative advantage, we obtain 95% confidence intervals by resampling 10th-grade schools, and Appendix Table B.6 reports how this share changes with the fixed effects, the definition of reading, and the construction of comparative advantage.

To separate math comparative advantage from absolute mathematics achievement, we also enter the two 10th-grade measures together, which amounts to letting the mathematics and reading scores have separate slopes. Following Gelbach (2016), we split the resulting reduction of the gap between the two measures: each part is the measure's slope in the joint model times its conditional gender difference. The split does not depend on the order in which the measures enter, but it does depend on the pair, so we also report it for the mathematics and reading scores.

Peer rank enters the model in the same way as the other measures, in the school-cohort cells in which it is defined (Section 2.1): first alone, then together with the mathematics score, as in the rank literature, and with a cubic in the score. Within a school-cohort cell, the rank is a monotone transformation of the score, so these models describe how the two measures are related to choices; we do not use variation in the composition of school cohorts to identify an effect of rank. Finally, we examine heterogeneity by interacting $Female_i$ with each performance measure and, separately, with quartiles of 10th-grade mathematics defined among all students at the end of compulsory education.

Three features limit interpretation. First, school fixed effects absorb time-invariant differences across schools, including course offerings, but not within-school differences in preferences, beliefs, or encouragement. Second, reaching each stage selects students differently by gender and achievement, so cross-stage gaps need not measure the same parameter. Among *Batxillerat* students of the 2000–2005 birth cohorts, 65.5% of girls and 60.0% of boys enter university. A Lee-style trimming exercise (Lee, 2009) bounds the raw STEM-degree gap between -36.6 and -27.5 percentage points (Appendix Table B.7), while inverse-probability reweighting on observed characteristics lowers the share of the gap accounted for by math comparative advantage from 20.0% to 16.9% (Appendix Table B.6). Third, university enrollment reflects both applications and admission.

Measurement error introduces further uncertainty. Comparative advantage combines errors from two correlated scores and may therefore be less reliable than either score, particularly when mathematics scores are concentrated near the ceiling. Rank also inherits error from the mathematics

score. Because no reliability estimates exist for these assessments, Appendix Table B.8 examines classical measurement error under assumed score reliabilities and error correlations. Under these assumptions, correcting the separately entered mathematics and comparative-advantage measures increases the shares of the gap they account for, while the split in the joint model changes substantially.

4 Results

All models follow Section 3 unless stated otherwise. Except for the estimates that use peer rank or the science score, the Science-track estimates use the 317,552 students observed in *Batxillerat* and the STEM-degree estimates the 147,726 university entrants who remain in the fixed-effect models. Unless another table is cited, every share below is in Appendix Table B.5.

Shares across the two margins. The Science-track gap is 16.3 percentage points. Adding 10th-grade mathematics achievement reduces it by 43.1%, to 9.2 points; adding comparative advantage reduces it by 49.4%, to 8.2 points. Appendix Table B.1 reports the full estimates. Reading alone widens the gap to 19.0 points because girls score higher and reading positively predicts enrollment when mathematics is omitted. Its coefficient turns negative once mathematics is included.

Among university entrants, the STEM-degree gap is 28.6 percentage points. Mathematics achievement reduces it by 16.5%, to 23.9 points, and comparative advantage by 20.0%, to 22.9 points; reading alone slightly widens it. Appendix Tables B.2 to B.4 report the full estimates for the three STEM definitions. Under the definition that also includes life sciences and medicine, the gap is smaller, 18.6 points, because girls in the Science track frequently enter health-related programs, whereas boys rarely do, and enrollment rises more steeply with comparative advantage (0.137 versus 0.108), so the respective shares are larger, 37.1% and 39.1%. That definition is the closer counterpart of the Science track, which prepares students for those fields too: its slope and its reduction of the gap, 0.137 and 7.3 points, are nearly those of the Science track, 0.138 and 8.0. The math-intensive results resemble those under the canonical definition. The accounting identity locates the contrast between the margins. Girls' conditional comparative advantage is 0.58 below boys' in *Batxillerat* and 0.53 below among university entrants; the corresponding slopes are 0.138 and 0.108. The reductions are therefore 8.0 and 5.7 percentage points, with the larger STEM-degree gap accounting for much of the difference in shares.

Measured in 6th grade, mathematics achievement and comparative advantage account for much less of both gaps: about 22% and 15% of the Science-track gap, and 9% and 6% of the STEM-degree gap (Appendix Tables B.1 and B.2). At both margins, every performance slope is larger in 10th grade. The larger 10th-grade Science-track share reflects both a steeper comparative-advantage slope (0.138 versus 0.071) and a larger conditional gender difference (0.58 versus 0.35;

the 6th-grade value is not shown).

Of the robustness checks in Appendix Table B.6, the construction of comparative advantage matters most: the shares of the two gaps it accounts for are 63.2% and 23.6% using the standardized raw-score difference z_{M-R}, and 44.4% and 18.8% using the percentile-rank difference.

Comparative advantage and mathematics achievement. The accounting identity also shows why comparative advantage accounts for more of the gap than mathematics achievement. Within *Batxillerat*, its slope is flatter (0.138 versus 0.208), but its conditional gender difference is larger (0.58 versus 0.34). Figure 3 shows, on a common sample, how much of each gap six 10th-grade measures close, or widen, when entered one at a time; their slopes are in Appendix Tables B.11 and B.12. At both margins, comparative advantage again has a flatter slope than mathematics achievement yet closes more of the gap, so girls and boys differ more in it. On this sample, the 10th-grade science score accounts for about half as much of each gap as mathematics achievement, 22.8% and 8.9% against 44.2% and 16.6%, mainly because girls and boys differ less in it: its slope is close to that of mathematics in the Science track and somewhat flatter for a STEM degree.

Entered jointly, 10th-grade mathematics achievement and comparative advantage reduce the Science-track gap by 46.8%, to 8.7 percentage points, and the STEM-degree gap by 20.2%, to 22.8 points.[10] At the same mathematics score, a higher reading score is associated with only slightly lower enrollment: the slopes are 0.215 on mathematics and −0.015 on reading in the Science track, and 0.153 and −0.032 for a STEM degree, so reading accounts for 4.7% and 10.7% of the joint reduction (Appendix Table B.5). The Gelbach (2016) decomposition splits the joint reduction between the two measures and attributes to comparative advantage 5.2% of the Science-track gap and 6.0% of the STEM-degree gap.

This split depends on parameterization and measurement error. Expressed in mathematics and reading scores, the same model attributes an even larger share of each gap to mathematics. Under the classical-error assumptions in Appendix Table B.8, setting both score reliabilities to 0.90 raises the share of the gap attributed to comparative advantage in the joint model from 5.2% and 6.0% to 23.0% and 16.4%, and to 8.7% and 8.4% if the errors correlate at 0.4. The observed scores therefore show substantial overlap between the two measures, while the precise split remains sensitive to measurement assumptions.

Peer rank. The same question of overlap arises for the peer comparison. On the rank sample (Section 2.1), rank remains positively associated with enrollment conditional on mathematics achievement: a one-standard-deviation increase in rank corresponds to 11.3 percentage points for the Science track and 8.6 for a STEM degree, while the mathematics slopes fall by about

[10] Comparative advantage alone accounts for more of the Science-track gap (49.4%) than the joint model that nests it because the share measures the change in the coefficient on girls, not model fit: comparative advantage alone values girls' reading advantage at its own slope, whereas the joint model gives reading almost none.

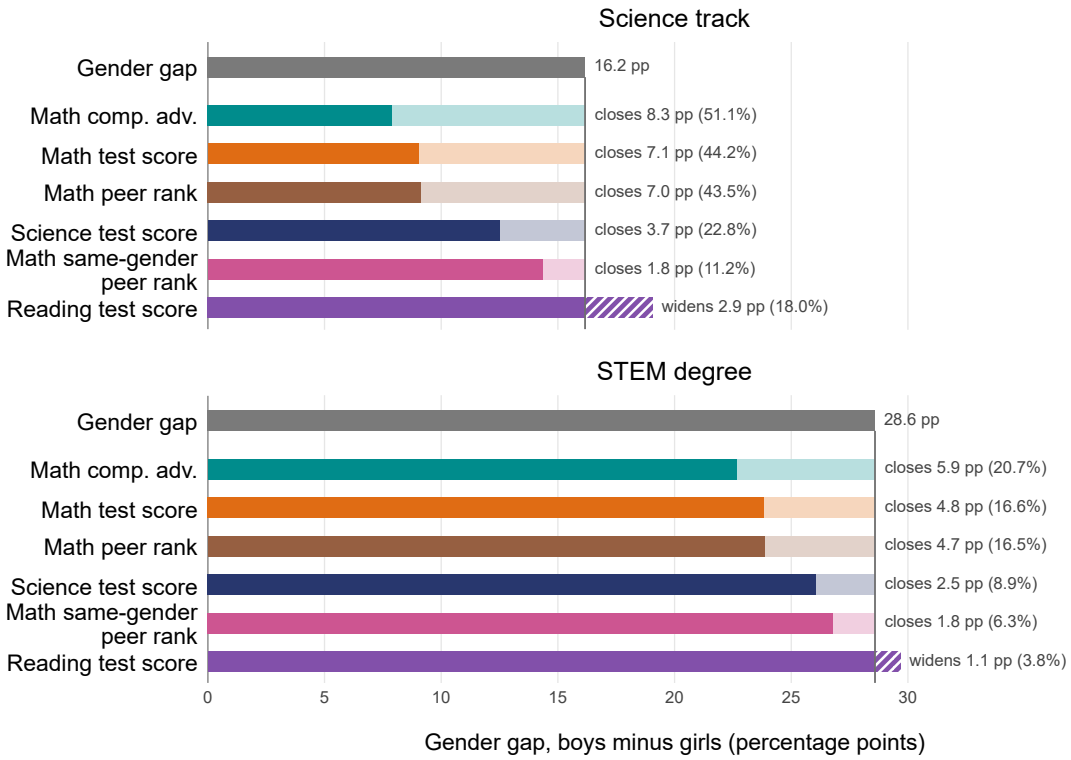


Figure 3: **Gender gap in Science-track and STEM-degree enrollment, with and without each 10th-grade performance measure.** Gender gap: the gap with the controls and the school and cohort fixed effects of equation (1) but no performance measure. Each bar below it is the gap once one measure is entered alone; the lighter segment is the part of the gender gap that the measure closes, and the hatched segment the part it adds. Labels: the change in percentage points and as a percentage of the gender gap. Common sample within each panel: *Batxillerat* students of the 2001–2007 birth cohorts with a science score, in school-cohort cells where more than 20 students at the end of compulsory education have one, at least 10 of them of the student's gender (N = 265,996), and the university entrants among them, of the 2001–2005 birth cohorts (N = 116,526). Scores are standardized within cohort and ranks over all cohorts, among the 371,419 students at the end of compulsory education with a science score in these cells. Source: Idescat administrative records; authors' calculations.

57% and 68%. Yet adding rank to mathematics barely changes the share of the gap accounted for, from 43.1% to 43.9% in the Science track and from 16.5% to 16.9% for a STEM degree, and a cubic in the mathematics score raises them only to 44.6% and 17.7% (Appendix Tables B.13 and B.14).

Entered alone, peer rank accounts for similar shares of the gaps as mathematics achievement. On the common sample of Figure 3, which requires a science score and so restricts the rank sample to the 2001–2007 birth cohorts, rank accounts for 43.5% of the Science-track gap and 16.5% of the STEM-degree gap, close to mathematics achievement's 44.2% and 16.6% (Appendix Tables B.11 and B.12). Same-gender rank accounts for only 11.2% and 6.3% of the two gaps despite a steep slope: ranking each gender separately largely removes the gender difference by construction, and what remains reflects which girls and boys reach each stage. Rank is constructed from the mathematics score and inherits its measurement error, limiting interpretation of the share each accounts for separately.

Where the gap remains. Most of the STEM-degree gap arises after the track choice. Track composition accounts for 28.0% of the raw 28.7-point gap under the symmetric decomposition (17.2–38.7% using girls' or boys' enrollment rates as reference), and 19.4 of the remaining 20.7 points arise within the Science track (Appendix Table B.9). Adding track indicators reduces the gap from 28.6 to 21.3 points, and among Science-track entrants, where the gap is 35.5 points, comparative advantage accounts for 11.7% of it, versus 20.0% among all entrants (Appendix Table B.10). The two margins also differ in how the gap that remains relates to academic profiles. The differences below are in percentage points; given girls' lower enrollment rates, they shrink or reverse on the latent-index scale of a probit. In the Science track, girls' and boys' slopes on comparative advantage are statistically indistinguishable at the 5% level (with z_{M-R}, girls' slope is lower by 0.020, standard error 0.002; Appendix Table B.6, Panel B). At university, girls' slope is a little more than half of boys' (0.083 versus 0.144). With STEM-degree enrollment rates of 12.8% for girls and 41.5% for boys (Appendix Table A.2, Panel D), however, the interaction on the latent-index scale is positive, both in a probit controlling for school-by-cohort regressor means (0.049, standard error 0.011) and in a conditional logit within school-cohort cells (0.170, standard error 0.020; Appendix Table B.15, Panel C).

The Science-track gap rises from 7.6 percentage points in the bottom mathematics quartile to 10.8 in the top; the STEM-degree gap rises from 10.0 to 31.2 points (Appendix Table B.15).[11] University entrants are concentrated in the upper quartiles, making bottom-quartile estimates less precise. On the latent-index scale, the STEM-degree gap changes only from -0.80 to -0.85 and the Science-track gap from -0.26 to -0.32 (Appendix Table B.15, Panel C). At both margins, these latent-index gaps are statistically indistinguishable across the bottom three quartiles; the top-quartile gap is larger than the bottom-quartile one in the Science track and, at university, larger than the third-quartile one but not statistically different from the bottom one. Latent-index gaps are comparable across quartiles within each margin, but not across margins.

5 Conclusion

This paper asks how two reference points for students' mathematics performance, their own performance in reading and their standing among schoolmates, relate to the gender gap in STEM specialization at the choice of the Science track and at the choice of a STEM degree.

Four results stand out. First, 10th-grade math comparative advantage accounts for about half of the gender gap in Science-track enrollment but about a fifth of the gap in STEM-degree enrollment (39.1% when STEM also includes life sciences and medicine), a contrast that comes mainly from the size of the STEM-degree gap; measured in 6th grade, it accounts for much less of

[11] Under the trimming assumptions in Appendix Table B.7, differential university entry does not appear to account for the widening under the canonical STEM definition: within quartiles, the bounds for the top math quartile lie entirely below those for the bottom one. Under the definition that also includes life sciences and medicine, they overlap.

both. Second, comparative advantage accounts for more than mathematics achievement because girls and boys differ more in it, and much of what it accounts for is shared with mathematics achievement; how the reduction splits between the two cannot be pinned down without the reliability of the assessments. Third, peer rank, constructed from the mathematics score, adds little to what that score accounts for and inherits its measurement error. Fourth, in percentage points, at the choice of track girls' and boys' enrollment rises at similar rates with comparative advantage and the gap changes relatively little across the mathematics distribution, whereas at university girls' enrollment rises a little more than half as steeply and the gap more than triples between the bottom and top quartiles of mathematics; on the latent-index scale, the widening largely disappears and girls' slope on comparative advantage is, if anything, the steeper.

These results have limits. They are associations on selected samples: each choice is observed only among students who reach that stage, more girls than boys reach *Batxillerat*, and university enrollment also reflects applications and admission. University outcomes rest on six birth cohorts, the two youngest observed only in their first years of possible entry, and on enrollment in Catalan universities. Measurement error makes the shares that mathematics achievement and comparative advantage account for lower bounds under classical error, and since a difference of two scores carries the error of both, the part the joint model attributes to comparative advantage could be several times larger at plausible reliabilities if the errors on the two tests are independent. Finally, the 6th- and 10th-grade assessments differ and the later one is closer to the decision, so the larger 10th-grade shares do not show that comparative advantage formed in secondary school matters more for choices.

What the results do show is where explanations beyond academic profiles are most needed. At the choice of track, differences in academic profiles account for a large part of the gender gap. At university, they account for only about a fifth of it: a girl and a boy from the same school and cohort, with similar family backgrounds and the same mathematics and reading scores, still differ by 22.8 percentage points in their probability of enrolling in a STEM degree. Whatever leads them to different degrees is not visible in their test scores.

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Appendix A. Additional Descriptive Evidence

Field of study	STEM degree (math-intensive)	STEM degree	STEM degree (including life sciences and medicine)
In all three definitions			
Chemistry	✓	✓	✓
Earth sciences	✓	✓	✓
Physics	✓	✓	✓
Mathematics	✓	✓	✓
Statistics	✓	✓	✓
Natural sciences, mathematics and statistics, not further defined	✓	✓	✓
Software and applications development and analysis	✓	✓	✓
Information and communication technologies, not elsewhere classified	✓	✓	✓
Information and communication technologies, not further defined	✓	✓	✓
Chemical engineering and processes	✓	✓	✓
Environmental protection technology	✓	✓	✓
Electricity and energy	✓	✓	✓
Electronics and automation	✓	✓	✓
Mechanics and metal trades	✓	✓	✓
Motor vehicles, ships and aircraft	✓	✓	✓
Engineering, not elsewhere classified	✓	✓	✓
Building and civil engineering	✓	✓	✓
Engineering, manufacturing and construction, not further defined	✓	✓	✓
Not in the math-intensive definition			
Environmental sciences	✗	✓	✓
Computer use	✗	✓	✓
Database and network design and administration	✗	✓	✓
Architecture and town planning	✗	✓	✓
Only in the definition including life sciences and medicine			
Biology	✗	✗	✓
Biochemistry	✗	✗	✓
Natural environments and wildlife	✗	✗	✓
Crop and livestock production	✗	✗	✓
Forestry	✗	✗	✓
Veterinary	✗	✗	✓
Dental studies	✗	✗	✓
Medicine	✗	✗	✓
Nursing and midwifery	✗	✗	✓
Therapy and rehabilitation	✗	✗	✓
Pharmacy	✗	✗	✓
Health, not elsewhere classified	✗	✗	✓

Table A.1: **Fields of study in each definition of a STEM degree.** Fields of the Catalan Classification of Education Fields (CCED 2020) included (✓) or not (✗) in each definition. Source: Idescat administrative records; authors' calculations.

	Girls	Boys	Difference (boys − girls)
Panel A. Student, family, and school characteristics			
Foreign-born student	0.07	0.06	-0.01^{***}
			(0.001)
Two-parent household	0.90	0.91	0.01^{***}
			(0.002)
Same-sex parents	0.013	0.012	0.000
			(0.0004)
All parents with a university degree	0.19	0.20	0.01^{**}
			(0.003)
One parent with a university degree	0.21	0.22	0.01^{***}
			(0.001)
One foreign-born parent	0.094	0.092	-0.002^{*}
			(0.0009)
Both parents foreign-born	0.114	0.109	-0.005^{***}
			(0.0011)
Public school, 10th grade	0.617	0.612	−0.005
			(0.0046)
Panel B. Scores in 6th grade			
6th grade Math test score	−0.10	0.11	0.21^{***}
			(0.004)
6th grade Math test score (0–100)	78.79	81.90	3.11^{***}
			(0.054)
6th grade Reading test score	0.09	−0.09	-0.18^{***}
			(0.004)
6th grade Reading test score (0–100)	77.58	75.23	-2.35^{***}
			(0.050)
6th grade Math comp. adv.	−0.19	0.20	0.39^{***}
			(0.003)
Panel C. Scores in 10th grade			
10th grade Math test score	−0.12	0.13	0.25^{***}
			(0.005)
10th grade Math test score (0–100)	66.15	71.01	4.86^{***}
			(0.092)
10th grade Reading test score	0.18	−0.19	-0.37^{***}
			(0.005)
10th grade Reading test score (0–100)	79.37	75.26	-4.11^{***}
			(0.052)
10th grade Math comp. adv.	−0.30	0.31	0.62^{***}
			(0.003)
Observations	224,878	219,155	
10th grade Science test score	−0.05	0.05	0.10^{***}
			(0.004)
10th grade Science test score (0–100)	66.16	67.84	1.68^{***}
			(0.072)
Observations	196,467	192,224	
Panel D. Continuation and STEM enrollment			
Continues to *Batxillerat* (%)	77.7	65.1	-12.6^{***}
			(0.28)
Science track (%)	40.8	57.5	16.7^{***}
			(0.25)
STEM degree (%)	12.8	41.5	28.7^{***}
			(0.27)
STEM degree, including life sciences and medicine (%)	31.8	50.7	18.9^{***}
			(0.32)
STEM degree, math-intensive (%)	10.4	38.6	28.2^{***}
			(0.26)

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.2: **Background, performance, and enrollment, by gender.** Students at the end of compulsory education. Panels A to C: proportions, except test scores, standardized within subject, grade, and cohort, or raw (0 to 100); public schools exclude publicly funded private (*concertada*) schools; science scores cover the 2001–2007 birth cohorts. Panel D: continuation to *Batxillerat* among all students at the end of compulsory education, the Science track among students observed in *Batxillerat*, and STEM degrees among university entrants of the 2000–2005 birth cohorts observed in *Batxillerat*. Source: Idescat administrative records; authors' calculations.

	Girls	Boys	All
Math test score	0.676	0.663	0.674
Reading test score	0.729	0.700	0.715
Math comp. adv.	0.334	0.341	0.388
Observations	224,878	219,155	444,033

Table A.3: **Correlation between 6th- and 10th-grade performance measures, within student.** Pearson correlations among students at the end of compulsory education. Source: Idescat administrative records; authors' calculations.

	Girls	Boys	Difference (boys − girls)	Normalized difference
10th grade Math test score	−0.12	0.13	0.25***	0.25
			(0.005)	
10th grade Math comp. adv.	−0.30	0.31	0.62***	0.75
			(0.003)	
10th grade Math peer rank	0.46	0.54	0.08***	0.26
			(0.001)	
Observations	215,267	209,212		

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table A.4: **Peer rank in math, math score, and comparative advantage, by gender.** Peer rank: percentile rank in 10th-grade math among the students of the same school and cohort, from 0 (lowest) to 1 (highest). Normalized difference: the difference divided by the square root of the average of the two variances. Students in school-cohort cells with more than 20 students and at least 10 of their own gender; their means differ from those of Appendix Table A.2 only in the third decimal. Source: Idescat administrative records; authors' calculations.

	Girls	Boys	All
Correlation between 10th grade Math peer rank and 10th grade Math comp. adv.	0.483	0.357	0.435
Share whose peer rank and comp. adv. disagree	0.327	0.330	0.329
of which: above the cohort median in Math, comp. adv. in Reading	0.211	0.118	0.165
of which: below the cohort median in Math, comp. adv. in Math	0.116	0.213	0.163
Observations	215,267	209,212	424,479

Table A.5: **Agreement between math comparative advantage and peer rank.** Students in school-cohort cells with more than 20 students and at least 10 of their own gender. Cohort median: the median of the student's school and cohort. Source: Idescat administrative records; authors' calculations.

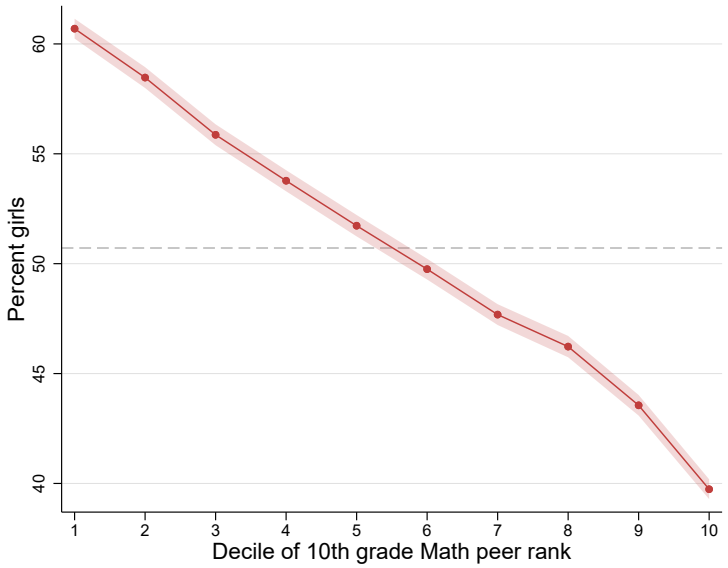


Figure A.1: **Share of girls by decile of peer rank in math.** Peer rank: percentile rank in 10th-grade math among the students at the end of compulsory education in the same school and cohort, in fixed intervals from [0, 0.1) to [0.9, 1]. 95% confidence intervals. The dashed line marks the share of girls in the sample (50.7%). Sample: students in school-cohort cells with more than 20 students at the end of compulsory education and at least 10 students of their own gender (N = 424,479). Source: Idescat administrative records; authors' calculations.

Appendix B. Additional Results

	Science track							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Girls	−0.167***	−0.163***	−0.127***	−0.170***	−0.138***	−0.0925***	−0.190***	−0.0824***
	(0.00252)	(0.00232)	(0.00228)	(0.00227)	(0.00234)	(0.00223)	(0.00223)	(0.00237)
6th grade Math test score			0.139***					
			(0.00120)					
6th grade Reading test score				0.0903***				
				(0.00131)				
6th grade Math comp. adv.					0.0707***			
					(0.00115)			
10th grade Math test score						0.208***		
						(0.00129)		
10th grade Reading test score							0.112***	
							(0.00157)	
10th grade Math comp. adv.								0.138***
								(0.00127)
Constant	0.575***	0.486***	0.461***	0.483***	0.476***	0.442***	0.487***	0.456***
	(0.00224)	(0.00381)	(0.00363)	(0.00373)	(0.00378)	(0.00347)	(0.00370)	(0.00370)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School and cohort FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	317,554	317,552	317,552	317,552	317,552	317,552	317,552	317,552
R-squared	0.028	0.054	0.108	0.074	0.065	0.172	0.081	0.098

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.1: **Science-track enrollment: gender gap and academic performance.** Linear probability models; students observed in *Batxillerat*, 2000–2007 birth cohorts. Scores in standard deviations; comparative advantage is their difference. Controls: student and parental foreign birth, household structure, and parental university education. Source: Idescat administrative records; authors' calculations.

	STEM degree							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Girls	−0.287***	−0.286***	−0.261***	−0.287***	−0.269***	−0.239***	−0.296***	−0.229***
	(0.00271)	(0.00269)	(0.00265)	(0.00267)	(0.00273)	(0.00263)	(0.00268)	(0.00276)
6th grade Math test score			0.0919***					
			(0.00133)					
6th grade Reading test score				0.0464***				
				(0.00156)				
6th grade Math comp. adv.					0.0550***			
					(0.00154)			
10th grade Math test score						0.139***		
						(0.00143)		
10th grade Reading test score							0.0519***	
							(0.00161)	
10th grade Math comp. adv.								0.108***
								(0.00144)
Constant	0.415***	0.375***	0.337***	0.361***	0.369***	0.302***	0.360***	0.349***
	(0.00263)	(0.00449)	(0.00441)	(0.00452)	(0.00444)	(0.00440)	(0.00448)	(0.00444)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School and cohort FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	147,732	147,726	147,726	147,726	147,726	147,726	147,726	147,726
R-squared	0.107	0.124	0.149	0.130	0.132	0.180	0.130	0.155

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.2: **STEM-degree enrollment: gender gap and academic performance.** Linear probability models; university entrants of the 2000–2005 birth cohorts observed in *Batxillerat*. Scores in standard deviations; comparative advantage is their difference. Controls: student and parental foreign birth, household structure, and parental university education. Source: Idescat administrative records; authors' calculations.

	STEM degree (including life sciences and medicine)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Girls	−0.189***	−0.186***	−0.151***	−0.188***	−0.167***	−0.117***	−0.205***	−0.113***
	(0.00315)	(0.00307)	(0.00302)	(0.00301)	(0.00309)	(0.00294)	(0.00298)	(0.00316)
6th grade Math test score			0.130***					
			(0.00160)					
6th grade Reading test score				0.0804***				
				(0.00186)				
6th grade Math comp. adv.					0.0637***			
					(0.00180)			
10th grade Math test score						0.203***		
						(0.00167)		
10th grade Reading test score							0.101***	
							(0.00198)	
10th grade Math comp. adv.								0.137***
								(0.00173)
Constant	0.507***	0.425***	0.372***	0.401***	0.419***	0.318***	0.396***	0.393***
	(0.00258)	(0.00542)	(0.00527)	(0.00541)	(0.00536)	(0.00510)	(0.00534)	(0.00527)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School and cohort FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	147,732	147,726	147,726	147,726	147,726	147,726	147,726	147,726
R-squared	0.036	0.058	0.095	0.071	0.066	0.150	0.075	0.096

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.3: **STEM-degree enrollment, definition including life sciences and medicine: gender gap and academic performance.** Linear probability models; university entrants of the 2000–2005 birth cohorts observed in *Batxillerat*. Scores in standard deviations; comparative advantage is their difference. Controls: student and parental foreign birth, household structure, and parental university education. Source: Idescat administrative records; authors' calculations.

	STEM degree (math-intensive)							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Girls	−0.282***	−0.281***	−0.259***	−0.283***	−0.267***	−0.238***	−0.291***	−0.231***
	(0.00260)	(0.00257)	(0.00252)	(0.00255)	(0.00260)	(0.00250)	(0.00256)	(0.00262)
6th grade Math test score			0.0845***					
			(0.00129)					
6th grade Reading test score				0.0448***				
				(0.00147)				
6th grade Math comp. adv.					0.0485***			
					(0.00146)			
10th grade Math test score						0.128***		
						(0.00139)		
10th grade Reading test score							0.0514***	
							(0.00158)	
10th grade Math comp. adv.								0.0958***
								(0.00138)
Constant	0.386***	0.352***	0.317***	0.338***	0.347***	0.285***	0.337***	0.329***
	(0.00257)	(0.00428)	(0.00422)	(0.00432)	(0.00423)	(0.00422)	(0.00429)	(0.00422)
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School and cohort FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	147,732	147,726	147,726	147,726	147,726	147,726	147,726	147,726
R-squared	0.112	0.129	0.151	0.134	0.135	0.179	0.135	0.155

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.4: **STEM-degree enrollment, math-intensive definition: gender gap and academic performance.** Linear probability models; university entrants of the 2000–2005 birth cohorts observed in *Batxillerat*. Scores in standard deviations; comparative advantage is their difference. Controls: student and parental foreign birth, household structure, and parental university education. Source: Idescat administrative records; authors' calculations.

	Science track	STEM degree	STEM degree (including life sciences and medicine)	STEM degree (math-intensive)
Panel A. Conditional gender difference in the measure, girls minus boys				
10th grade Math test score	−0.337	−0.338	−0.338	−0.338
10th grade Reading test score	0.245	0.192	0.192	0.192
10th grade Math comp. adv.	−0.582	−0.530	−0.530	−0.530
Panel B. Share of the gap accounted for (%)				
10th grade Math test score	43.1	16.5	37.1	15.4
10th grade Reading test score	−16.9	−3.5	−10.4	−3.5
10th grade Math comp. adv.	49.4	20.0	39.1	18.1
95 percent interval	[47.8, 50.8]	[19.3, 20.7]	[37.4, 40.8]	[17.5, 18.7]
Panel C. Math and comp. adv. entered jointly (10th grade)				
Coefficient on girls	−0.0866	−0.228	−0.112	−0.230
Share of the gap accounted for (%)	46.8	20.2	39.6	18.3
Split of the reduction (%)				
Math test score	88.8	70.6	89.8	74.8
Math comp. adv.	11.2	29.4	10.2	25.2
95 percent interval	[8.8, 13.1]	[26.6, 32.0]	[7.3, 12.8]	[22.4, 28.1]
Same model, split between Math and Reading (%)				
Math test score	95.3	89.3	96.3	90.9
Reading test score	4.7	10.7	3.7	9.1
Same model, slopes on the two scores				
Math test score	0.215	0.153	0.209	0.138
Reading test score	−0.015	−0.032	−0.014	−0.024
Gap attributed to Math comp. adv. (%)	5.2	6.0	4.1	4.6
Controls	Yes	Yes	Yes	Yes
School and cohort FE	Yes	Yes	Yes	Yes
Observations	317,552	147,726	147,726	147,726

Table B.5: **Accounting for the gender gap in STEM enrollment, 10th-grade measures.** Linear probability models: the Science track among students observed in *Batxillerat*; STEM degrees among university entrants of the 2000–2005 birth cohorts observed in *Batxillerat*. Panel C splits the reduction of the gap between the two measures, following Gelbach (2016). Intervals: 2.5th and 97.5th percentiles of a cluster bootstrap over 10th-grade schools. Source: Idescat administrative records; authors' calculations.

	Science track	STEM degree	STEM degree (including life sciences and medicine)	STEM degree (math-intensive)
Panel A. Share of the gap accounted for by 10th grade Math comp. adv. (%)				
Main specification	49.4	20.0	39.1	18.1
School × cohort FE, common sample	51.1	20.8	40.6	18.7
Reading as the higher language score	46.9	18.8	37.1	17.0
Math comp. adv. as z_{M-R}	63.2	23.6	49.6	21.6
Math comp. adv. as a percentile-rank difference	44.4	18.8	35.5	16.9
Reweighted for selection into university		16.9	30.5	14.7
Panel B. Girls × 10th grade Math comp. adv.				
Math comp. adv. as $z_M - z_R$	0.00434	-0.0609^{***}	-0.00860^{*}	-0.0690^{***}
	(0.00229)	(0.00305)	(0.00345)	(0.00306)
Math comp. adv. as z_{M-R}	-0.0198^{***}	-0.0944^{***}	-0.0403^{***}	-0.101^{***}
	(0.00180)	(0.00251)	(0.00285)	(0.00250)
Observations, main specification	317,552	147,726	147,726	147,726

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.6: **Robustness of the main results.** Panel A: each row after the first changes one element of the main specification. School × cohort FE: students common to both fixed-effect structures, on whom additive fixed effects give the shares of the first row. Higher language score: the higher of the Catalan and Spanish scores. Percentile-rank difference: the difference between the within-cohort percentile ranks of the math and reading scores. Reweighted: university entrants weighted by the inverse of their probability of entering university, from a logit of entry on gender, the 10th-grade math score and comparative advantage, the controls, and cohort, estimated on the students observed in *Batxillerat*. Source: Idescat administrative records; authors' calculations.

	Gender gap among university entrants, girls − boys (percentage points)			
	Raw gap	Lower bound	Upper bound	Entrants
Panel A. **STEM degree**				
All entrants	−28.7	−36.6	−27.5	147,732
		(0.39)	(0.24)	
10th grade Math quartile 1	−10.0	−12.7	−8.5	12,865
		(0.63)	(0.68)	
10th grade Math quartile 2	−14.6	−20.4	−12.4	28,670
		(0.43)	(0.49)	
10th grade Math quartile 3	−22.7	−33.9	−20.1	44,405
		(0.35)	(0.43)	
10th grade Math quartile 4	−31.2	−36.8	−29.5	61,792
		(0.48)	(0.39)	
Trimming within math quartiles		−32.6	−23.4	
		(0.29)	(0.26)	
Panel B. **STEM degree (including life sciences and medicine)**				
All entrants	−18.9	−25.1	−16.0	147,732
		(0.36)	(0.29)	
10th grade Math quartile 1	−6.6	−18.4	−0.4	12,865
		(0.73)	(0.94)	
10th grade Math quartile 2	−9.3	−27.2	−2.3	28,670
		(0.47)	(0.65)	
10th grade Math quartile 3	−12.4	−28.5	−5.6	44,405
		(0.71)	(0.54)	
10th grade Math quartile 4	−14.2	−17.8	−10.5	61,792
		(0.45)	(0.45)	
Trimming within math quartiles		−22.3	−7.5	
		(0.35)	(0.32)	
Panel C. **STEM degree (math-intensive)**				
All entrants	−28.2	−36.4	−27.2	147,732
		(0.39)	(0.23)	
10th grade Math quartile 1	−9.3	−11.3	−8.2	12,865
		(0.60)	(0.64)	
10th grade Math quartile 2	−13.8	−18.3	−12.1	28,670
		(0.41)	(0.46)	
10th grade Math quartile 3	−22.2	−30.9	−20.2	44,405
		(0.34)	(0.41)	
10th grade Math quartile 4	−31.7	−37.6	−30.3	61,792
		(0.48)	(0.38)	
Trimming within math quartiles		−31.7	−23.8	
		(0.28)	(0.25)	

Standard errors of the bounds in parentheses: analytic (Lee 2009), not clustered
Entering university: 65.5 percent of girls and 60.0 percent of boys in *Batxillerat*

Table B.7: **STEM-degree gap and selection into university: Lee bounds.** Raw gender gap among university entrants of the 2000–2005 birth cohorts observed in *Batxillerat*, overall and by quartile of the 10th-grade math score, with the bounds of Lee (2009). They trim the excess share of girls among entrants from either end of girls' outcomes, assuming that no student who would enter university as a boy would fail to enter as a girl and that gender is otherwise independent of the outcome. Trimming within math quartiles describes the data rather than bounding the gap, since the quartile is not independent of gender. Source: Idescat administrative records; authors' calculations.

Reliability of both scores	Science track			STEM degree		
	Math, alone (%)	Math comp. adv., alone (%)	Math comp. adv., with Math (%)	Math, alone (%)	Math comp. adv., alone (%)	Math comp. adv., with Math (%)
1.00	43.1	49.4	5.2	16.5	20.0	6.0
0.95	46.5	59.5	11.9	18.2	25.0	9.7
0.90	50.5	75.0	23.0	20.2	33.4	16.4
0.90, errors correlated at 0.4	50.5	62.1	8.7	20.2	26.3	8.4
Controls		Yes			Yes	
School and cohort FE		Yes			Yes	
Observations		317,552			147,726	

Table B.8: **Gender gaps in STEM enrollment: the shares under assumed reliabilities of the test scores.** Shares of the gender gap with the slopes of the 10th-grade math and reading scores corrected for classical measurement error, at a common reliability of the two scores and with independent errors except in the last row. Math comparative advantage, with Math: the part of the reduction of the gap attributed to comparative advantage when the math score and comparative advantage enter jointly, following Gelbach (2016), as a share of the gap. Reliability: the assumed share of each score's within-cohort variance that is not error. The corrected share of comparative advantage alone exceeds 100 percent below a common reliability of about 0.85 in the Science track and 0.80 for a STEM degree, and the correction is undefined below 0.71 and 0.75. Source: Idescat administrative records; authors' calculations.

	STEM degree	STEM degree (including life sciences and medicine)	STEM degree (math-intensive)
Panel A. The gap and its three terms (percentage points)			
Gap, boys minus girls	28.7***	18.9***	28.2***
	(0.27)	(0.32)	(0.26)
Track composition, at the STEM rates of girls	4.9***	12.2***	4.1***
	(0.10)	(0.22)	(0.09)
STEM rates within tracks, at the track shares of girls	17.6***	5.4***	17.7***
	(0.20)	(0.19)	(0.20)
Interaction	6.2***	1.3***	6.4***
	(0.15)	(0.09)	(0.15)
Panel B. Share of the gap due to track composition (%)			
STEM rates of girls as reference	17.2***	64.5***	14.4***
	(0.33)	(0.89)	(0.29)
STEM rates of boys as reference	38.7***	71.3***	37.0***
	(0.59)	(0.84)	(0.58)
Average of the two (Kitagawa)	28.0***	67.9***	25.7***
	(0.44)	(0.84)	(0.41)
Panel C. Kitagawa within-track term, by track (percentage points)			
Within-track term, total	20.7***	6.1***	20.9***
	(0.21)	(0.21)	(0.21)
Science track	19.4***	5.0***	19.8***
	(0.21)	(0.18)	(0.20)
Humanities and social sciences track	1.1***	0.8***	1.0***
	(0.07)	(0.08)	(0.06)
Arts or general track	0.2***	0.2***	0.2***
	(0.05)	(0.05)	(0.05)
Observations	147,732	147,732	147,732
10th-grade schools	1,108	1,108	1,108
Controls	No	No	No
School and cohort FE	No	No	No

Boys minus girls, raw means. Delta-method standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.9: **Where the STEM-degree gap arises: *Batxillerat* track composition or choice within the track.** Raw gender gap in STEM-degree enrollment among university entrants of the 2000–2005 birth cohorts observed in *Batxillerat*, split into the difference in track shares (composition), the difference in enrollment rates within tracks, and their interaction. Panel B values the difference in track shares at girls' rates, at boys' rates, or at their average (Kitagawa, 1955). Panel C splits the Kitagawa within-track term by track. Source: Idescat administrative records; authors' calculations.

	STEM degree			
	All entrants		Science-track entrants	
	(1)	(2)	(3)	(4)
Girls	-0.286^{***}	-0.213^{***}	-0.355^{***}	-0.314^{***}
	(0.00269)	(0.00225)	(0.00354)	(0.00373)
10th grade Math comp. adv.				0.0920^{***}
				(0.00242)
Reduction of the gap from the previous column (%)		25.4		11.7
Track indicators	No	Yes	No	No
Controls	Yes	Yes	Yes	Yes
School and cohort FE	Yes	Yes	Yes	Yes
Observations	147,726	147,726	78,605	78,605

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.10: **STEM-degree enrollment: gender gap and *Batxillerat* track.** Linear probability models; university entrants of the 2000–2005 birth cohorts observed in *Batxillerat*. Track indicators: Humanities and social sciences, and Arts or general, with Science as the reference. Source: Idescat administrative records; authors' calculations.

	Science track							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Girls	−0.166***	−0.162***	−0.0902***	−0.191***	−0.125***	−0.0790***	−0.0913***	−0.143***
	(0.00269)	(0.00250)	(0.00240)	(0.00240)	(0.00235)	(0.00254)	(0.00238)	(0.00244)
10th grade Math test score			0.210***					
			(0.00134)					
10th grade Reading test score				0.113***				
				(0.00160)				
10th grade Science test score					0.196***			
					(0.00134)			
10th grade Math comp. adv.						0.138***		
						(0.00137)		
10th grade Math peer rank (standardized)							0.191***	
							(0.00119)	
10th grade Math same-gender peer rank (standardized)								0.184***
								(0.00118)
Constant	0.576***	0.485***	0.443***	0.488***	0.458***	0.453***	0.454***	0.481***
	(0.00239)	(0.00423)	(0.00385)	(0.00413)	(0.00394)	(0.00408)	(0.00382)	(0.00388)
Share of the gap accounted for (%)			44.2	−18.0	22.8	51.1	43.5	11.2
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School and cohort FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	265,996	265,996	265,996	265,996	265,996	265,996	265,996	265,996
R-squared	0.027	0.055	0.175	0.083	0.164	0.100	0.177	0.171

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.11: **Science-track enrollment: gender gap, academic performance, and peer rank.** Linear probability models; students observed in *Batxillerat*, 2001–2007 birth cohorts, with a 10th-grade science score, in school-cohort cells where more than 20 students at the end of compulsory education have one, at least 10 of them of the student's gender. Scores are standardized within cohort and ranks over all cohorts, among all students with a science score in these cells. Source: Idescat administrative records; authors' calculations.

	STEM degree							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Girls	−0.288***	−0.286***	−0.238***	−0.297***	−0.261***	−0.227***	−0.239***	−0.268***
	(0.00301)	(0.00303)	(0.00300)	(0.00301)	(0.00297)	(0.00315)	(0.00299)	(0.00299)
10th grade Math test score			0.142***					
			(0.00161)					
10th grade Reading test score				0.0522***				
				(0.00182)				
10th grade Science test score					0.119***			
					(0.00156)			
10th grade Math comp. adv.						0.109***		
						(0.00166)		
10th grade Math peer rank (standardized)							0.124***	
							(0.00141)	
10th grade Math same-gender peer rank (standardized)								0.121***
								(0.00141)
Constant	0.421***	0.378***	0.303***	0.363***	0.327***	0.351***	0.321***	0.337***
	(0.00285)	(0.00521)	(0.00508)	(0.00521)	(0.00509)	(0.00511)	(0.00505)	(0.00505)
Share of the gap accounted for (%)			16.6	−3.8	8.9	20.7	16.5	6.3
Controls	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
School and cohort FE	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	116,526	116,526	116,526	116,526	116,526	116,526	116,526	116,526
R-squared	0.106	0.125	0.181	0.131	0.170	0.156	0.184	0.182

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.12: **STEM-degree enrollment: gender gap, academic performance, and peer rank.** Linear probability models; university entrants of the 2001–2005 birth cohorts observed in *Batxillerat*, with a 10th-grade science score, in school-cohort cells where more than 20 students at the end of compulsory education have one, at least 10 of them of the student's gender. Scores are standardized within cohort and ranks over all cohorts, among all students with a science score in these cells. Source: Idescat administrative records; authors' calculations.

	Science track				
	(1)	(2)	(3)	(4)	(5)
Girls	−0.163***	−0.0927***	−0.0936***	−0.0914***	−0.0903***
	(0.00238)	(0.00229)	(0.00227)	(0.00228)	(0.00227)
10th grade Math test score		0.209***		0.0904***	0.125***
		(0.00131)		(0.00373)	(0.00453)
10th grade Math peer rank (standardized)			0.190***	0.113***	0.0747***
			(0.00114)	(0.00349)	(0.00373)
Constant	0.485***	0.441***	0.454***	0.448***	0.415***
	(0.00392)	(0.00356)	(0.00353)	(0.00354)	(0.00368)
Share of the gap accounted for (%)	0.0	43.1	42.6	43.9	44.6
Cubic in Math test score	No	No	No	No	Yes
Controls	Yes	Yes	Yes	Yes	Yes
School and cohort FE	Yes	Yes	Yes	Yes	Yes
Observations	304,459	304,459	304,459	304,459	304,459
R-squared	0.054	0.172	0.174	0.177	0.181

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.13: **Science-track enrollment: gender gap, math score, and peer rank.** Linear probability models; students observed in *Batxillerat*, 2000–2007 birth cohorts, in school-cohort cells with more than 20 students at the end of compulsory education and at least 10 of their own gender. Scores and ranks in standard deviations. Source: Idescat administrative records; authors' calculations.

	STEM degree				
	(1)	(2)	(3)	(4)	(5)
Girls	−0.286***	−0.239***	−0.239***	−0.238***	−0.236***
	(0.00278)	(0.00271)	(0.00272)	(0.00271)	(0.00269)
10th grade Math test score		0.140***		0.0440***	0.0538***
		(0.00147)		(0.00435)	(0.00481)
10th grade Math peer rank (standardized)			0.122***	0.0864***	0.0419***
			(0.00129)	(0.00398)	(0.00398)
Constant	0.375***	0.302***	0.320***	0.313***	0.274***
	(0.00460)	(0.00450)	(0.00445)	(0.00449)	(0.00465)
Share of the gap accounted for (%)	0.0	16.5	16.4	16.9	17.7
Cubic in Math test score	No	No	No	No	Yes
Controls	Yes	Yes	Yes	Yes	Yes
School and cohort FE	Yes	Yes	Yes	Yes	Yes
Observations	141,699	141,699	141,699	141,699	141,699
R-squared	0.125	0.181	0.183	0.184	0.191

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.14: **STEM-degree enrollment: gender gap, math score, and peer rank.** Linear probability models; university entrants of the 2000–2005 birth cohorts observed in *Batxillerat*, in school-cohort cells with more than 20 students at the end of compulsory education and at least 10 of their own gender. Scores and ranks in standard deviations. Source: Idescat administrative records; authors' calculations.

	Science track	STEM degree
Panel A. Slope of the measure, by gender		
10th grade Math test score		
Boys	0.226^{***}	0.217^{***}
	(0.00180)	(0.00257)
Girls	0.196^{***}	0.0980^{***}
	(0.00144)	(0.00145)
Girls − boys	-0.0296^{***}	-0.119^{***}
	(0.00190)	(0.00281)
10th grade Reading test score		
Boys	0.103^{***}	0.0632^{***}
	(0.00196)	(0.00284)
Girls	0.121^{***}	0.0422^{***}
	(0.00198)	(0.00182)
Girls − boys	0.0176^{***}	-0.0209^{***}
	(0.00236)	(0.00343)
10th grade Math comp. adv.		
Boys	0.136^{***}	0.144^{***}
	(0.00179)	(0.00268)
Girls	0.140^{***}	0.0830^{***}
	(0.00163)	(0.00150)
Girls − boys	0.00434	-0.0609^{***}
	(0.00229)	(0.00305)
Observations	317,552	147,726
Panel B. Gender gap within each quartile of the 10th grade math score		
Quartile 1 (lowest)	-0.0756^{***}	-0.0999^{***}
	(0.00427)	(0.00635)
Quartile 2	-0.0872^{***}	-0.146^{***}
	(0.00388)	(0.00483)
Quartile 3	-0.105^{***}	-0.227^{***}
	(0.00368)	(0.00437)
Quartile 4 (highest)	-0.108^{***}	-0.312^{***}
	(0.00321)	(0.00388)
p-value: equal gender gap across quartiles	<0.001	<0.001
Observations	317,552	147,726
Panel C. Latent-index scale		
Girls × 10th grade Math comp. adv., probit	0.0215^{**}	0.0486^{***}
	(0.00669)	(0.0110)
Girls × 10th grade Math comp. adv., conditional logit	0.0374^{***}	0.170^{***}
	(0.0113)	(0.0197)
Gender gap, quartile 1 (lowest)	-0.259^{***}	-0.797^{***}
	(0.0140)	(0.0385)
Gender gap, quartile 2	-0.240^{***}	-0.759^{***}
	(0.0105)	(0.0219)
Gender gap, quartile 3	-0.266^{***}	-0.808^{***}
	(0.00938)	(0.0153)
Gender gap, quartile 4 (highest)	-0.317^{***}	-0.851^{***}
	(0.00945)	(0.0114)
p-value: equal gap, quartiles 1 to 3	0.129	0.157
p-value: quartile 4 equals quartile 1	<0.001	0.168
p-value: quartile 4 equals quartile 3	<0.001	0.018
Observations, probit	317,554	147,732
Observations, conditional logit	317,105	145,502
Controls	Yes	Yes
School and cohort FE, Panels A and B	Yes	Yes

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table B.15: **Slopes by gender and the gender gap by math quartile.** Linear probability models: the Science track among students observed in *Batxillerat*; STEM degrees among university entrants of the 2000–2005 birth cohorts observed in *Batxillerat*. Panel A: one model per measure, interacting it with the indicator for girls. Panel B: one model interacting the quartile with the indicator for girls; quartiles are defined among all students at the end of compulsory education. Panel C: probit with cohort effects, the family controls, and the school-by-cohort means of the regressors in place of school fixed effects; conditional logit within school-cohort cells, which drops cells with no variation in the outcome; the quartile gaps come from one probit. Source: Idescat administrative records; authors' calculations.

Appendix C. Results for All Students at the End of Compulsory Education

This appendix reports the models of Section 4 for all students at the end of compulsory education, where the outcomes do not condition on enrollment: a student who never enters *Batxillerat* counts as not enrolled in the Science track, and a student of the 2000–2005 birth cohorts who does not enter a Catalan university counts as not enrolled in a STEM degree (Appendix Table C.1). Each outcome combines two decisions, continuing beyond compulsory schooling and choosing STEM at the stage reached, so these estimates describe overall STEM participation rather than field choice among students who continue. Like the core analysis of Breda and Napp (2019), these models relate scores measured around age 15 to an outcome defined for all students; their outcome, however, is a stated intention rather than realized enrollment, so the two sets of estimates are not directly comparable.

With controls and fixed effects, girls are 5.5 percentage points less likely than boys to enroll in the Science track, against 16.3 points among students observed in *Batxillerat*. Adding 10th-grade mathematics achievement reduces the gap to 0.6 points, 89.5% less, and adding math comparative advantage to 0.7 points, 87.3% less, while the reading score about doubles it, to 10.9 points. In the 2000–2005 birth cohorts, girls are 10.0 percentage points less likely to enroll in a STEM degree. Mathematics achievement reduces the gap by 20.4%, to 8.0 points, and comparative advantage by 22.7%, to 7.7 points, somewhat above the 16.5% and 20.0% among university entrants. Under the definition that also includes life sciences and medicine, on a gap of 3.6 points, the shares are 83.6% and 85.3%, and under the math-intensive definition 18.3% and 19.8%.

	Unconditional on post-compulsory education enrollment			
	Science track	**STEM degree**	**STEM degree (including life sciences and medicine)**	**STEM degree (math-intensive)**
Share of boys enrolled	0.374	0.169	0.207	0.157
Observations	444,032	324,311	324,311	324,311
Panel A. Coefficient on girls				
No controls or fixed effects	-0.0573^{***}	-0.102^{***}	-0.0379^{***}	-0.103^{***}
	(0.00233)	(0.00193)	(0.00185)	(0.00185)
Controls and fixed effects only	-0.0548^{***}	-0.1000^{***}	-0.0358^{***}	-0.101^{***}
	(0.00189)	(0.00182)	(0.00157)	(0.00176)
With 6th grade Math test score	-0.0263^{***}	-0.0871^{***}	-0.0170^{***}	-0.0896^{***}
	(0.00183)	(0.00174)	(0.00151)	(0.00169)
With 6th grade Reading test score	-0.0789^{***}	-0.110^{***}	-0.0504^{***}	-0.110^{***}
	(0.00184)	(0.00185)	(0.00159)	(0.00180)
With 6th grade Math comp. adv.	-0.0430^{***}	-0.0956^{***}	-0.0306^{***}	-0.0976^{***}
	(0.00188)	(0.00176)	(0.00156)	(0.00172)
With 10th grade Math test score	-0.00573^{**}	-0.0795^{***}	-0.00587^{***}	-0.0827^{***}
	(0.00174)	(0.00169)	(0.00145)	(0.00164)
With 10th grade Reading test score	-0.109^{***}	-0.125^{***}	-0.0732^{***}	-0.124^{***}
	(0.00179)	(0.00191)	(0.00163)	(0.00186)
With 10th grade Math comp. adv.	-0.00695^{***}	-0.0772^{***}	-0.00525^{***}	-0.0812^{***}
	(0.00190)	(0.00167)	(0.00157)	(0.00162)
Panel B. Slope of the measure				
6th grade Math test score	0.143^{***}	0.0628^{***}	0.0918^{***}	0.0570^{***}
6th grade Reading test score	0.126^{***}	0.0565^{***}	0.0850^{***}	0.0518^{***}
6th grade Math comp. adv.	0.0301^{***}	0.0116^{***}	0.0139^{***}	0.00988^{***}
10th grade Math test score	0.205^{***}	0.0931^{***}	0.136^{***}	0.0846^{***}
10th grade Reading test score	0.144^{***}	0.0641^{***}	0.0975^{***}	0.0591^{***}
10th grade Math comp. adv.	0.0776^{***}	0.0377^{***}	0.0506^{***}	0.0332^{***}
Panel C. Conditional gender difference in the measure (girls minus boys)				
6th grade Math test score	-0.200	-0.204	-0.204	-0.204
6th grade Reading test score	0.192	0.172	0.172	0.172
6th grade Math comp. adv.	-0.392	-0.377	-0.377	-0.377
10th grade Math test score	-0.239	-0.220	-0.220	-0.220
10th grade Reading test score	0.377	0.384	0.384	0.384
10th grade Math comp. adv.	-0.616	-0.603	-0.603	-0.603
Panel D. Share of the gap accounted for (%)				
6th grade Math test score	52.0	12.8	52.5	11.5
6th grade Reading test score	-44.0	-9.7	-40.9	-8.8
6th grade Math comp. adv.	21.5	4.4	14.6	3.7
10th grade Math test score	89.5	20.4	83.6	18.3
10th grade Reading test score	-99.2	-24.6	-104.5	-22.4
10th grade Math comp. adv.	87.3	22.7	85.3	19.8
Controls	Yes	Yes	Yes	Yes
School and cohort FE	Yes	Yes	Yes	Yes
Observations	444,031	324,306	324,306	324,306

Standard errors, clustered by 10th-grade school, in parentheses
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table C.1: **Accounting for the gender gap in STEM enrollment among all students at the end of compulsory education.** Linear probability models with outcomes that do not condition on enrollment: the Science track among all students at the end of compulsory education, and STEM degrees among those of the 2000–2005 birth cohorts. Source: Idescat administrative records; authors' calculations.